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Hamiltonian squares in digraphs under semi-degree conditions☆

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ABSTRACT

We study the square of a Hamilton cycle in digraphs under a minimum semi-degree condition. Motivated by a question of DeBiasio et al. [Combin. Probab. Comput. 35 (2026)], we show that for every $\gamma > 0$, every sufficiently large n-vertex digraph D satisfying $\delta^0(D) \geq \left(\frac{3}{4}+\gamma\right)n$ contains the square of a Hamilton cycle. This provides a partial answer, for the case $k = 2$, to the question raised in their work, and may be viewed as a directed analogue of the classical theorem of Faudree et al. [Recent Adv. Graph Theory (1991)] for undirected graphs. The total-degree theorem of DeBiasio et al. immediately yields a sufficient semi-degree bound of $4n/5 + o(n)$, and our result improves this bound to $3n/4 + o(n)$.

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1. Introduction

Hamiltonicity under degree conditions is a central theme in extremal graph theory. A classical theorem of Dirac [4] asserts that every graph G on $n \geq 3$ vertices with minimum degree $\delta(G)$ at least $n/2$ contains a Hamilton cycle. For digraphs, a theorem of Ghouila-Houri [12] states that every strongly connected n-vertex digraph D with minimum total degree $\delta(D)$ at least n contains a directed Hamilton cycle. In particular, a minimum semi-degree $\delta^0(D) \geq n/2$ already guarantees a directed Hamilton cycle. These foundational results naturally lead to the study of stronger spanning structures under analogous degree conditions.

A natural strengthening of Hamiltonicity is the existence of powers of a Hamilton cycle. For a directed cycle $C = v_1v_2\cdots v_tv_1$, its *kth power* is obtained by adding the arc v_iv_j whenever the directed distance from v_i to v_j along C is at most k (with indices taken modulo t). The 2nd power is usually called the *square* of the cycle.

For undirected graphs, Pósa and Seymour [6,22] conjectured that the minimum degree of at least $kn/(k+1)$ forces the kth power of a Hamilton cycle in every n-vertex graph. This problem stimulated a long line of work. After a series of further contributions, see [7–11,16,17], Komlós et al. [18] confirmed the Pósa–Seymour conjecture for all sufficiently large graphs. Hamiltonian-type theorems have also found applications beyond Hamiltonicity itself. Katona [14] developed constructions based on Hamiltonian-type theorems, illustrating the usefulness of Hamiltonian ideas as tools in extremal combinatorics.

The directed setting, however, is substantially more delicate. Even for basic Hamilton cycles in oriented graphs (that is, orientations of simple graphs), finding the correct semi-degree threshold was a longstanding problem until Keevash et al. [15] proved that $\delta^0(D) \geq (3n-4)/8$ is sufficient and best possible for large oriented graphs. See [19] for a survey

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including many of the results on the background of Hamilton cycles in digraphs, oriented graphs and tournaments. For powers of Hamilton cycles in directed and oriented graphs, much less was known until very recently. For tournaments, Draganić et al. [5] proved the tight minimum semi-degree bound $\delta^0(T) \geq n/4 + cn^{1-1/\lceil k/2 \rceil}$ for the existence of the kth power of a Hamilton cycle. Moving from tournaments to oriented graphs, DeBiasio et al. [3] proved that the minimum semi-degree condition of $\delta^0(D) \geq (1/2 - 1/10^{6000k})n$ suffices to guarantee the kth power of a Hamilton cycle in any sufficiently large oriented graph D.

For general digraphs, in the same paper [3], they also made a major breakthrough by asymptotically determining the minimum total degree threshold forcing the square of a Hamilton cycle. More precisely, for every $\eta > 0$, every sufficiently large n-vertex digraph D with $\delta(D) \geq (8/5 + \eta)\, n$ contains the square of a Hamilton cycle. Since $\delta(D) \geq 2\delta^0(D)$, this result immediately implies that, for every $\varepsilon > 0$, the condition $\delta^0(D) \geq (4/5 + \varepsilon)n$ is sufficient.

Closely related to the square of a cycle, the *reverse square* of a cycle C is the digraph with the same vertex set as C, whose arc set is the union of all arcs of C and all arcs yx for which the directed distance from x to y in C is exactly 2. Zhang et al. [24] proved that every sufficiently large digraph D with $\delta^0(D) \geq (2/3 + \gamma)\, n$ contains the reverse square of a Hamilton cycle. Although the reverse square and the square share the same Hamiltonian backbone, the latter is more restrictive, as the distance-two arcs must all be oriented consistently forward along the underlying cyclic order.

By contrast, determining the minimum semi-degree condition for the square of a Hamilton cycle in general digraphs appears to be a considerably more difficult problem. This problem was initially raised by Treglown [23] and later formalized by DeBiasio et al. [3] into the following question, which remains open even for $k = 2$.

Question 1.1 ([3]). *Does every n-vertex digraph D with $\delta^0(D) \geq \frac{k}{k+1}n$ contain the kth power of a Hamilton cycle?*

In this paper, we make progress on Question 1.1 in the case $k = 2$ by proving the following asymptotic result.

Theorem 1.2. *For every $\gamma > 0$, there exists $n_0 \in \mathbb{N}$ such that for any $n \geq n_0$ the following holds. Let D be an n-vertex digraph with*

$$\delta^0(D) \geq \left(\frac{3}{4} + \gamma\right) n.$$

Then D contains the square of a Hamilton cycle.

The total degree theorem of DeBiasio et al. [3] yields an asymptotic sufficient semi-degree bound of $4n/5$, and Theorem 1.2 improves this bound to $3n/4$. Our result may also be viewed as a directed minimum semi-degree analogue of the classical theorem of Faudree et al. [11] for undirected graphs, which was an important step toward the resolution of the Pósa–Seymour conjecture for undirected graphs.

Theorem 1.3 ([11]). *For every $\varepsilon > 0$, there exists a constant $c = c(\varepsilon)$ such that every graph G on n vertices with*

$$\delta(G) \geq \left(\frac{3}{4} + \varepsilon\right) n + c$$

contains the square of a Hamilton cycle.

As a further consequence of Theorem 1.2, we derive the following factor theorem for the square of a cycle. Here, C_ℓ^2 denotes the square of a cycle on ℓ vertices, and by a C_ℓ^2-factor we mean a spanning subgraph consisting of pairwise vertex-disjoint copies of C_ℓ^2.

Corollary 1.4. *For every $\gamma > 0$, there exists $\ell_0 = \ell_0(\gamma)$ such that for all $\ell \geq \ell_0$ and every integer $m \geq 1$, any digraph D on ℓm vertices with minimum semi-degree*

$$\delta^0(D) \geq \left(\frac{3}{4} + \gamma\right) \ell m$$

contains a C_ℓ^2-factor.

Organization. The paper is organized as follows. In Section 2, we introduce the necessary notation, probabilistic tools, and outline our proof strategy. In Section 3, we establish the almost-covering, absorbing, and connecting lemmas. Finally, in Section 4, we combine these ingredients to prove Theorem 1.2 and subsequently deduce Corollary 1.4.

2. Preliminaries

2.1. Notation

For notation not defined in this paper, we refer the reader to [1]. Throughout the paper, all digraphs are finite and loopless. For a digraph D, we write $V(D)$ and $A(D)$ for its vertex set and arc set, respectively, and let $|D| = |V(D)|$. Given a

positive integer t, we write $[t] := \{1, \dots, t\}$. For a set X, its cardinality is denoted by $|X|$. If $X \subseteq V(D)$, then $D[X]$ denotes the subdigraph of D induced by X, and we write $D - X = D[V(D) \setminus X]$. More generally, if H is a subdigraph of D, then $D - H = D[V(D) \setminus V(H)]$.

For a vertex $v \in V(D)$, let $N_D^+(v) = \{u \in V(D) : vu \in A(D)\}$ and $N_D^-(v) = \{u \in V(D) : uv \in A(D)\}$ be the *out-neighborhood* and *in-neighborhood* of v, respectively. Their sizes are denoted by $d_D^+(v) = |N_D^+(v)|$ and $d_D^-(v) = |N_D^-(v)|$. For a set $X \subseteq V(D)$, we further write $N_D^+(v, X) = N_D^+(v) \cap X$, $N_D^-(v, X) = N_D^-(v) \cap X$, $d_D^+(v, X) = |N_D^+(v, X)|$, and $d_D^-(v, X) = |N_D^-(v, X)|$. For vertices $x_1, \dots, x_t \in V(D)$, we use the shorthand $N_D^+(x_1, \dots, x_t) = \bigcap_{i=1}^t N_D^+(x_i)$ and $N_D^-(x_1, \dots, x_t) = \bigcap_{i=1}^t N_D^-(x_i)$. The *minimum out-degree, minimum in-degree, minimum semi-degree, and minimum total degree* of D are denoted by $\delta^+(D)$, $\delta^-(D)$, $\delta^0(D) = \min\{\delta^+(D), \delta^-(D)\}$, and $\delta(D) = \min_{v \in V(D)}(d_D^+(v) + d_D^-(v))$, respectively. Whenever the underlying digraph is clear from the context, we omit the subscript D.

All paths and cycles in digraphs are directed unless stated otherwise. For a directed path $Q = v_1 v_2 \cdots v_\ell$, its kth power is obtained by adding the arcs $v_i v_j$ for all $1 \le i < j \le \ell$ with $j - i \le k$. For a directed cycle $C = v_1 v_2 \cdots v_\ell v_1$, its kth power is obtained by adding the arcs $v_i v_{i+r}$ for every $i \in [\ell]$ and every $r \in [k]$, where the subscripts are interpreted modulo ℓ. We refer to the kth power of a path or cycle as a k-path or k-cycle, respectively. In particular, a 2-*path* is the square of a path. If $P = v_1 v_2 \cdots v_\ell$ is a 2-path with $\ell \ge 4$, then we call $v_1 v_2$ and $v_{\ell-1} v_\ell$ the initial arc and the terminal arc of P, respectively.

Finally, we use the standard hierarchy notation $0 < 1/n \ll a \ll b \ll c \ll 1$ to mean that the constants are chosen from right to left, each sufficiently small as a function of those to its right. All asymptotic statements are understood for sufficiently large n.

2.2. Proof strategy

We briefly outline the proof of Theorem 1.2 in four main steps, building upon the connecting–absorbing method of Rödl, Ruciński, and Szemerédi [21].

Step 1: Constructing an absorbing 2-path. We first find a short 2-path P_A with the property that any sufficiently small leftover vertex set can be absorbed into P_A while preserving its end-arcs.

Step 2: Choosing a reservoir. Next we choose a vertex set $\mathcal{R}$ at random such that, for any two prescribed disjoint arcs and any small forbidden set, one can find a short 2-path inside $\mathcal{R}$ connecting these arcs. This reservoir will later be used to join the 2-path obtained from the covering step.

Step 3: Covering almost all remaining vertices. After deleting $V(P_A) \cup \mathcal{R}$, we apply an almost covering lemma to the remaining digraph. This yields a bounded collection of pairwise vertex-disjoint 2-paths $\mathcal{P}$ covering all but a small exceptional set. Here we use the result of DeBiasio et al. [3] on almost covering by k-paths under a minimum total degree condition, together with the simple observation that $\delta(D) \ge 2\delta^0(D)$.

Step 4: Connecting and final absorbing. Finally, we connect the absorbing path and the bounded family of 2-paths $\mathcal{P}$ by using the reservoir set. In this way, we obtain a large 2-cycle covering all but a very small number of vertices. The absorbing property of P_A then incorporates these remaining vertices, completing the square of a Hamilton cycle.

2.3. Probabilistic tools

In our proofs, we shall repeatedly use the following standard probabilistic tools.

Lemma 2.1 ([13]). *Let X be a random variable drawn from either a binomial or a hypergeometric distribution, with expectation $\mathbb{E}X$. Let a be any real number with $0 < a < 1$. Then the following statements hold.*

(1) (Chernoff bound)

$$\mathbb{P}\big(X \le (1-a)\mathbb{E}X\big) \le e^{-\frac{a^2}{2}\mathbb{E}X},$$

and

$$\mathbb{P}\big(X \ge (1+a)\mathbb{E}X\big) \le e^{-\frac{a^2}{3}\mathbb{E}X}.$$

(2) (Markov's inequality) *For any non-negative random variable Y and any $t > 0$,*

$$\mathbb{P}(Y \ge t) \le \frac{\mathbb{E}Y}{t}.$$

3. Almost covering, absorbing, and connecting lemmas for Theorem 1.2

3.1. Almost covering lemma

We shall use Lemma 5.2 of DeBiasio et al. [3].

Lemma 3.1 ([3]). *For every integer $k \geq 2$ and every $\eta > 0$, there exist $n_0, T \in \mathbb{N}$ such that, for every $n \geq n_0$, the following holds. If G is an n-vertex digraph satisfying $\delta(G) \geq 2(1 - \frac{1}{k+2} + \eta)n$, then G contains a collection of at most T vertex-disjoint k-paths covering all but at most ηn vertices.*

Since $\delta(D) \geq 2\delta^0(D)$, this immediately yields the following minimum semi-degree version.

Lemma 3.2 (*Almost Covering Lemma*). *For any integer $k \geq 2$ and real number $\gamma > 0$, there exist $n_0 = n_0(k, \gamma) \in \mathbb{N}$ and $T = T(k, \gamma) \in \mathbb{N}$ such that, for every integer $n \geq n_0$, the following holds. If D is an n-vertex digraph with $\delta^0(D) \geq (1-\frac{1}{k+2}+\gamma)n$, then D contains a collection of at most T disjoint k-paths that cover all but at most γn vertices in D.*

3.2. Absorbing lemma

The proof of the main theorem relies on the absorbing method. In this section, we prove Lemma 3.4, which guarantees the existence of many small absorbing structures for each vertex, and will later allow us to incorporate a small leftover set into the kth power of a Hamilton cycle. The following definition is a crucial notion needed for constructing an absorbing k-path.

Definition 3.3. For a digraph D and a vertex $v \in V(D)$, we say a k-path on $2(k+2)$ vertices $P_v = v_1 \cdots v_{2(k+2)}$ in D is a $(k+2)$-absorber for v if $v_1 \cdots v_{k+2} v v_{k+3} \cdots v_{2(k+2)}$ is also a k-path in D.

Now we shall establish the absorbing lemma in a general form valid for every integer $k \geq 2$.

Lemma 3.4 (*Absorbing Lemma*). *For every integer $k \geq 2$ and every $\gamma > 0$, there exist $\xi = \xi(k, \gamma) > 0$ and $n_0 = n_0(k, \gamma)$ such that the following holds for every $n \geq n_0$. If D is an n-vertex digraph with $\delta^0(D) \geq (1 - \frac{1}{k+2} + \gamma)n$, then for every $v \in V(D)$, there are at least $\xi n^{2(k+2)}$ absorbers P_v for v in D.*

Proof. By the lower bound on $\delta^0(D)$, we have $d_D^-(v) \geq (1 - \frac{1}{k+2} + \gamma)n$. For any vertex $v_{k+2} \in N_D^-(v)$, we can get $|N_D^-(v_{k+2}) \cap N_D^-(v)| \geq 2\delta^-(D) - n \geq (1 - \frac{2}{k+2} + 2\gamma)n$. More generally, for every $j \in [k+2]$, after $v_{j+1}, \ldots, v_{k+2}$ have been chosen, we have

$$\begin{aligned}
&|N_D^-(v_{j+1}, \ldots, v_{k+2}, v) \setminus \{v_{j+1}, \ldots, v_{k+2}, v\}| \\
&\quad \geq (k-j+3)\left(1 - \frac{1}{k+2} + \gamma\right)n - (k-j+2)n - (k-j+3) \\
&\quad = \left(\frac{j-1}{k+2} + (k-j+3)\gamma\right)n - (k-j+3) \\
&\quad \geq \frac{1}{2}(k-j+3)\gamma n,
\end{aligned}$$

where the last inequality holds for sufficiently large n. Here an empty list of vertices is omitted when $j = k+2$. Consequently, we may choose $v_{k+2}, v_{k+1}, \ldots, v_1$ successively so that $v_1 v_2 \cdots v_{k+2} v$ is a k-path, with at least $\frac{1}{2}(k-j+3)\gamma n$ choices at the step corresponding to v_j. Moreover, for any $j' \in \{k+3, \ldots, 2(k+2)\}$, we have

$$\begin{aligned}
&|N_D^+(v_{j'-k}, \ldots, v_{j'-1}, v) \setminus \{v_1, \ldots, v_{j'-1}, v\}| \\
&\geq (k+1) \cdot \left(1 - \frac{1}{k+2} + \gamma\right)n - kn - j' \geq \frac{(k+1)\gamma n}{2}.
\end{aligned}$$

In summary, the number of absorbers $P_v = v_1 \cdots v_{k+2} v_{k+3} \cdots v_{2(k+2)}$ for v is at least

$$\left(\frac{(k+1)\gamma n}{2}\right)^{k+2} \prod_{j=1}^{k+2} \frac{(k-j+3)\gamma n}{2} \geq \left(\frac{\gamma}{2}\right)^{2(k+2)} n^{2(k+2)}.$$

Thus the assertion holds with $\xi = (\gamma/2)^{2(k+2)} > 0$. □

3.3. Connecting lemma

To connect disjoint arcs while avoiding prescribed vertices, we shall present the following lemma.

Lemma 3.5 (*Connecting Lemma*). *For any $0 < 1/n \ll \gamma < 1$, if D is an n-vertex digraph with $\delta^0(D) \geq \left(\frac{3}{4} + \gamma\right)n$, then for every pair of disjoint arcs $uv, yz \in A(D)$, and every set $W \subseteq V(D) \setminus \{u, v, y, z\}$ of size at most $\gamma n/2$, there exists a 2-path $x_1 x_2 \cdots x_6$ in $D \setminus W$ such that $x_1 = u$, $x_2 = v$, $x_5 = y$, and $x_6 = z$.*

Proof. Let $D' = D \setminus W$. Clearly, $\delta^0(D') \geq (3/4+\gamma)n - \gamma n/2 = (3/4+\gamma/2)n$. Define

$$U_1 := N^+_{D'}(u) \cap N^+_{D'}(v) \cap N^-_{D'}(y) \quad \text{and} \quad U_2 := N^+_{D'}(v) \cap N^-_{D'}(y) \cap N^-_{D'}(z).$$

Since $\delta^0(D') \geq (\frac{3}{4}+\gamma/2)n$, we have

$$|U_1|, |U_2| \geq 3\left(\frac{3}{4}+\frac{\gamma}{2}\right)n - 2n = \left(\frac{1}{4}+\frac{3\gamma}{2}\right)n.$$

In particular, we may choose $x_3 \in U_1 \setminus \{u, v, y, z\}$. It remains to choose a vertex $x_4 \in U_2 \cap N^+_{D'}(x_3)$. Using again the lower bound on $\delta^0(D')$, we obtain $d^+_{D'}(x_3) \geq (3/4+\gamma/2)n$. Hence

$$\begin{aligned} |U_2 \cap N^+_{D'}(x_3) \setminus \{x_3, u, v, y, z\}| &\geq |U_2| + |N^+_{D'}(x_3)| - n - 5 \\ &\geq \left(\frac{1}{4}+\frac{3\gamma}{2}\right)n + \left(\frac{3}{4}+\frac{\gamma}{2}\right)n - n - 5 \\ &= 2\gamma n - 5 \geq \gamma n. \end{aligned}$$

So we may pick $x_4 \in (U_2 \cap N^+_{D'}(x_3)) \setminus \{x_3, u, v, y, z\}$ with at least γn choices. Thus $x_1x_2x_3x_4x_5x_6 = uvx_3x_4yz$ is a 2-path on 6 vertices in $D \setminus W$ that starts with the arc uv and ends with the arc yz, as desired. □

4. Proofs of Theorem 1.2 and Corollary 1.4

4.1. Proof of Theorem 1.2

Choose constants $\eta, \xi > 0$ and $n_0, T \in \mathbb{N}$ such that

$$0 < 1/n_0 \ll 1/T \ll \eta \ll \xi \ll \gamma < 1.$$

Let $n \geq n_0$, and let D be an n-vertex digraph with

$$\delta^0(D) \geq \left(\frac{3}{4}+\gamma\right)n.$$

We divide the proof of Theorem 1.2 into four steps, as outlined in Section 2.2.

Step 1: Constructing an absorbing 2-path.

We first construct a 2-path P_A in D whose initial and terminal arcs are denoted by x_0y_0 and u_0v_0, respectively, and which satisfies the following properties:

(i) $|V(P_A)| \leq \eta^2 n$;
(ii) for every set $L \subseteq V(D - P_A)$ with $|L| \leq \eta^4 n$, the digraph $D[L \cup V(P_A)]$ contains a spanning 2-path from x_0y_0 to u_0v_0.

We first show (i). For each $v \in V(D)$, let

$$\mathcal{A}_v := \{P : P \text{ is a 4-absorber for } v\},$$

and write $\mathcal{A} := \bigcup_{v \in V(D)} \mathcal{A}_v$. Applying Lemma 3.4 with $k = 2$, we obtain

$$|\mathcal{A}_v| \geq \xi n^8 \qquad \text{for every } v \in V(D).$$

Since every 4-absorber uses 8 vertices, trivially $|\mathcal{A}| \leq n^8$.

Now let $\mathcal{C} \subseteq \mathcal{A}$ be a random collection obtained by including each element of $\mathcal{A}$ independently with probability $p = 4\eta^{8/3}n/|\mathcal{A}|$. Then

$$\mathbb{E}(|\mathcal{C}|) = p|\mathcal{A}| = 4\eta^{8/3}n.$$

Hence, by Lemma 2.1(2),

$$\mathbb{P}\big(|\mathcal{C}| \geq 8\eta^{8/3}n\big) \leq \frac{1}{2}. \tag{4.1}$$

Next fix $v \in V(D)$. Then

$$\mathbb{E}(|\mathcal{C} \cap \mathcal{A}_v|) = p|\mathcal{A}_v| \geq \frac{4\eta^{8/3}n}{|\mathcal{A}|} \cdot \xi n^8 \geq 4\eta^{8/3}n \cdot \xi \geq 4\eta^{10/3}n,$$

where the second inequality uses $|\mathcal{A}| \leq n^8$, and the last follows from $\eta \ll \xi$. Therefore, by Lemma 2.1(1) and the union bound,

$$\mathbb{P}\big(\exists v \in V(D) : |\mathcal{C} \cap \mathcal{A}_v| \leq 2\eta^{10/3}n\big) \leq 2n\exp(-\eta^{10/3}n/3) < \frac{1}{4}. \tag{4.2}$$

We also need to control intersections among the chosen absorbers. Let X be the number of ordered pairs of distinct intersecting absorbers in $\mathcal{C}$. Since each 4-absorber has 8 vertices, the number of ordered pairs of intersecting 4-absorbers is at most $64n^{15}$. Consequently,

$$\mathbb{E}(X) \le 64p^2n^{15} = \frac{1024\eta^{16/3}n^{17}}{|\mathcal{A}|^2} \le \frac{1024\eta^{16/3}}{\xi^2}\, n \le \eta^4 n,$$

where we used $|\mathcal{A}| \ge \xi n^8$. Again by Lemma 2.1(2) we obtain

$$\mathbb{P}(X \ge 4\eta^4 n) \le \frac{1}{4}. \tag{4.3}$$

Combining (4.1), (4.2), and (4.3), there is a choice of $\mathcal{C} \subseteq \mathcal{A}$ such that

$$|\mathcal{C}| \le 8\eta^{8/3}n, \qquad X \le 4\eta^4 n, \qquad \text{and} \qquad |\mathcal{C} \cap \mathcal{A}_v| \ge 2\eta^{10/3}n \text{ for all } v \in V(D).$$

Deleting from $\mathcal{C}$ at most $X \le 4\eta^4 n$ elements, we obtain a family $\mathcal{P} \subseteq \mathcal{A}$ of pairwise vertex-disjoint absorbers satisfying

$$|\mathcal{P}| \le 8\eta^{8/3}n \quad \text{and} \quad |\mathcal{P} \cap \mathcal{A}_v| \ge 2\eta^{10/3}n - 4\eta^4 n \ge \eta^{10/3}n \text{ for each } v \in V(D). \tag{4.4}$$

Write $\mathcal{P} = \{P_1, \dots, P_r\}$ with $r \le 8\eta^{8/3}n$. We now connect these absorbers into a single 2-path by greedily applying Lemma 3.5. Suppose that $P_1, \dots, P_i$ have already been connected into a single 2-path. Let $u_i v_i$ be the terminal arc of the resulting 2-path, and let $x_{i+1}y_{i+1}$ be the initial arc of P_{i+1}. These two arcs are vertex-disjoint since the absorbers in $\mathcal{P}$ are pairwise vertex-disjoint. Let F_i consist of all vertices in the absorbers together with the internal vertices of the connections already chosen, and set $W_i := F_i \setminus \{u_i, v_i, x_{i+1}, y_{i+1}\}$.

Then

$$|W_i| \le |F_i| \le 8r + 2(i-1) \le 10r \le 80\eta^{8/3}n \le \frac{\gamma n}{3} < \frac{\gamma n}{2}.$$

Hence Lemma 3.5 applies to the arcs $u_i v_i$ and $x_{i+1}y_{i+1}$ with forbidden set W_i. It yields a 2-path connecting these two arcs whose two internal vertices lie outside F_i.

Repeating this process for $i = 1, \dots, r-1$, we obtain a 2-path P_A containing all absorbers in $\mathcal{P}$. Moreover,

$$|V(P_A)| \le 8r + 2(r-1) \le 10r \le 80\eta^{8/3}n \le \eta^2 n.$$

This proves (i).

It remains to verify (ii). Let $L \subseteq V(D - P_A)$ with $|L| \le \eta^4 n$. For each $v \in L$, (4.4) gives at least $\eta^{10/3}n$ absorbers in $\mathcal{P} \cap \mathcal{A}_v$, and $\eta^{10/3}n \ge \eta^4 n$. Hence we may greedily assign to each $v \in L$ a distinct 4-absorber from $\mathcal{P} \cap \mathcal{A}_v$. Since these absorbers are pairwise vertex-disjoint and lie on P_A, we can absorb the vertices of L one by one inside P_A, preserving the same initial and terminal arcs. Therefore $D[L \cup V(P_A)]$ contains a spanning 2-path from x_0y_0 to u_0v_0, as required.

Step 2: Choosing a reservoir.

We next select a reservoir set $\mathcal{R} \subseteq V(D - P_A)$ satisfying:

(a) $|\mathcal{R}| = \eta^6 n$;
(b) given any vertex-disjoint arcs $ab, yz \in A(D)$ and any set $W \subseteq \mathcal{R} \setminus \{a, b, y, z\}$ of size at most $\eta^8 n$, there exists a 2-path on 6 vertices from ab to yz in $D[\mathcal{R} \cup \{a, b, y, z\}]$ which avoids W.

Choose uniformly at random a set $S \subseteq V(D - P_A)$ with $|S| = \eta^6 n$. For every vertex $v \in V(D)$, we have

$$d_D^{\pm}(v, V(D) \setminus V(P_A)) \ge \delta^0(D) - |V(P_A)| \ge \left(\frac{3}{4} + \gamma\right) n - \eta^2 n \ge \left(\frac{3}{4} + \frac{\gamma}{2}\right)\big(n - |V(P_A)|\big).$$

Therefore

$$\mathbb{E}\big(|N_D^{\pm}(v, S)|\big) = \frac{|S|}{n - |V(P_A)|}\, d_D^{\pm}(v, V(D) \setminus V(P_A)) \ge \left(\frac{3}{4} + \frac{\gamma}{2}\right)|S|.$$

Applying Lemma 2.1(1) for the hypergeometric distribution with $a = \frac{2\gamma}{9+6\gamma}$, we obtain

$$\mathbb{P}\left(|N_D^{\pm}(v) \cap S| \le \left(\frac{3}{4} + \frac{\gamma}{3}\right)|S|\right) \le \exp(-c|S|),$$

where $c = \frac{\gamma^2}{18(3+2\gamma)}$. Using the union bound we have

$$\mathbb{P}\left(\exists\, v \in V(D),\ \exists \star \in \{+, -\} : |N_D^{\star}(v) \cap S| \le \left(\frac{3}{4} + \frac{\gamma}{3}\right)|S|\right) \le 2n \exp\left(-\frac{\gamma^2}{18(3+2\gamma)}|S|\right) < \frac{1}{2}.$$

Hence there exists a set $\mathcal{R} \subseteq V(D) \setminus V(P_A)$ with $|\mathcal{R}| = \eta^6 n$ such that

$$|N_D^{\pm}(v) \cap \mathcal{R}| \ge \left(\frac{3}{4} + \frac{\gamma}{3}\right)|\mathcal{R}| \quad \text{for all } v \in V(D).$$

In particular,

$$\delta^0(D[\mathcal{R}]) \geq \left(\frac{3}{4}+\frac{\gamma}{3}\right)|\mathcal{R}|.$$

Indeed, property (b) follows from Lemma 3.5. To see this, consider the induced subgraph $R' := D[\mathcal{R}\cup\{a,b,y,z\}]$. Since $|R'| \leq |\mathcal{R}|+4$, we have

$$\delta^0(R') \geq \left(\frac{3}{4}+\frac{\gamma}{3}\right)|\mathcal{R}| \geq \left(\frac{3}{4}+\frac{\gamma}{4}\right)(|\mathcal{R}|+4) \geq \left(\frac{3}{4}+\frac{\gamma}{4}\right)|R'|,$$

where the second inequality follows from $\gamma|\mathcal{R}|/12 \geq 3+\gamma$, which holds for sufficiently large n because $|\mathcal{R}| = \eta^6 n$. We may therefore apply Lemma 3.5 to R' with $\gamma' = \gamma/4$. Moreover,

$$|W| \leq \eta^8 n = \eta^2|\mathcal{R}| \leq \frac{\gamma}{8}|\mathcal{R}| \leq \frac{\gamma'}{2}|R'|,$$

where $\eta^2 \leq \gamma/8$. Thus, Lemma 3.5 guarantees the existence of a 2-path on 6 vertices connecting ab and yz in R' that avoids W.

Step 3: Covering almost all remaining vertices.

In this step, we shall cover almost all vertices in D with a bounded family of vertex-disjoint 2-paths. Set $D' := D-(V(P_A)\cup\mathcal{R})$ and $|D'| := n'$. Since $|V(P_A)| \leq \eta^2 n$ and $|\mathcal{R}| = \eta^6 n$, we have

$$\delta^0(D') \geq \left(\frac{3}{4}+\gamma-\eta^2-\eta^6\right)n \geq \left(\frac{3}{4}+\eta^5\right)n'.$$

Applying Lemma 3.2 to D' with $k=2$ and with η^5 in place of γ, we obtain a family of at most T pairwise vertex-disjoint 2-paths in D' covering all but at most $\eta^5 n$ vertices of D'. Let these paths be $Q_1,\dots,Q_s$, where $s \leq T$.

Step 4: Connecting and final absorbing.

We will construct the desired square of Hamilton cycle in this step. For convenience, we set $Q_0 := P_A$. Recall that the initial and terminal arcs of Q_0 are x_0y_0 and u_0v_0, respectively. Furthermore, for each $i \in [s]$, let x_iy_i and u_iv_i denote the initial and terminal arcs of the 2-path Q_i obtained in Step 3. We now sequentially connect these 2-paths into a single 2-cycle using the reservoir $\mathcal{R}$.

Specifically, for each $i \in \{0,1,\dots,s\}$ (with indices taken modulo $s+1$, such that $Q_{s+1} = Q_0$), we aim to connect the terminal arc u_iv_i of Q_i to the initial arc $x_{i+1}y_{i+1}$ of Q_{i+1}. By property (b) of the reservoir in Step 2, there exists a 2-path R_i on 6 vertices connecting u_iv_i to $x_{i+1}y_{i+1}$. Each such connection utilizes exactly 2 internal vertices from $\mathcal{R}$. At the jth step of this process, the set W of previously used reservoir vertices has size $2j \leq 2(s+1)$. The constants T and η are fixed before n is chosen, so by enlarging n_0 if necessary we may assume that $2(T+1) \leq \eta^8 n$. Hence $|W| \leq 2(T+1) \leq \eta^8 n$, as required by property (b). In this way we obtain a 2-cycle C containing P_A and all the paths $Q_1,\dots,Q_s$.

Now we handle the remaining uncovered vertices. Let $L := V(D)\setminus V(C)$ and note that

$$|L| \leq \eta^5 n + |\mathcal{R}| = (\eta^5+\eta^6)n \leq \eta^4 n.$$

Moreover, by construction, $L \subseteq V(D-P_A)$. Therefore, the absorbing property (ii) in Step 1 applies: the induced subgraph $D[L\cup V(P_A)]$ contains a spanning 2-path Q having the same initial arc x_0y_0 and terminal arc u_0v_0 as P_A.

Finally, replacing the subpath P_A of C by Q, we obtain the square of a Hamilton cycle in D. This completes the proof of Theorem 1.2. □

4.2. Proof of Corollary 1.4

To prove Corollary 1.4, we combine Theorem 1.2 with the following equipartition lemma of Lo [20].

Lemma 4.1 ([20]). *There exists $\ell_p \in \mathbb{N}$ such that for all $\ell \geq \ell_p$ the following holds. Let $\delta > 0$ and $m \in \mathbb{N}$. Let G be a digraph on ℓm vertices with $\delta^0(G) \geq \delta\ell m$. Then there exists an equipartition of $V(G)$ into $V_1,\dots,V_m$ such that, for all $i \in [m]$, $|V_i| = \ell$ and*

$$\delta^0(G[V_i]) \geq \left(\delta - 10\ell^{-1/3}\right)\ell.$$

Proof of Corollary 1.4. Let n_0 be given by Theorem 1.2 with parameter $\gamma/2$, and let ℓ_p be the constant supplied by Lemma 4.1. Define

$$\ell_0(\gamma) := \max\{\ell_p, \lceil 20^3\gamma^{-3}\rceil, n_0\}.$$

Let D, ℓ, and m be as in the statement of Corollary 1.4.

Applying Lemma 4.1 to D, there exists an equipartition of $V(D)$ into $V_1, \dots, V_m$ such that, for all $i \in [m]$, $|V_i| = \ell$ and

$$\delta^0(D[V_i]) \geq \left(\frac{3}{4} + \gamma - 10\ell^{-1/3}\right)\ell \geq \left(\frac{3}{4} + \frac{\gamma}{2}\right)\ell.$$

Here the last inequality follows from $\ell \geq 20^3\gamma^{-3}$.

Since $\ell \geq n_0$, Theorem 1.2 implies that each $D[V_i]$ contains the square of a Hamilton cycle, that is, a spanning copy of C_ℓ^2. Therefore, D contains m vertex-disjoint copies of C_ℓ^2, one inside each V_i. Hence D contains a C_ℓ^2-factor. □

5. Concluding remarks

In this paper, we proved that, for every $\gamma > 0$, every sufficiently large n-vertex digraph D with $\delta^0(D) \geq (3/4 + \gamma)n$ contains the square of a Hamilton cycle. The minimum total degree result of DeBiasio et al. [3] yields an asymptotic sufficient semi-degree bound of $4n/5$, while our result lowers this bound to $3n/4$. Our theorem may also be viewed as a directed minimum semi-degree analogue of the classical result of Faudree et al. [11], and gives a partial answer, for $k = 2$, to the question raised in [3].

We conclude with a few remarks regarding potential avenues for future research.

The exact semi-degree threshold. A central open problem is to determine the exact minimum semi-degree threshold for the square of a Hamilton cycle in digraphs. In view of Question 1.1, it is natural to ask whether the correct threshold is $2n/3$.

Question 5.1. *Does every digraph on n vertices with $\delta^0(D) \geq \frac{2}{3}n$ contain the square of a Hamilton cycle?*

A standard extremal construction shows that this bound, if true, would be best possible. Indeed, let $V(D)$ be partitioned into an independent set S of size $\lfloor n/3 \rfloor + 1$ and a complete digraph K of size $\lceil 2n/3 \rceil - 1$, with all possible arcs between S and K. Then $\delta^0(D) = \lceil 2n/3 \rceil - 1$. However, D does not contain the square of a Hamilton cycle since along the underlying cyclic order of such a cycle, any two vertices of S must be separated by at least two vertices of K, implying $|K| \geq 2|S|$, a contradiction. Bridging the gap between our $(3/4 + \gamma)n$ bound and the conjectured $2n/3$ threshold, if possible, appears to be a highly nontrivial problem.

Ore degree conditions. Beyond Dirac-type conditions, it would be very interesting to establish an Ore-type analogue for the square of a Hamilton cycle in digraphs. For undirected graphs, Châu [2] proved that the mixed conditions $d(x) + d(y) \geq \frac{4}{3}n - \frac{1}{3}$ for every pair of non-adjacent vertices x, y and $\delta(G) > \frac{1}{3}n + 2$ guarantee the existence of the square of a Hamilton cycle. Châu also conjectured a generalization to the kth power of Hamilton cycles, supported by an extremal example. Motivated by this result, as well as by Question 1.1, we propose the following directed analogue.

Conjecture 5.2. *Let D be an n-vertex digraph and $k \geq 2$ be an integer. Suppose that $d^+(x) + d^-(y) \geq \frac{2k}{k+1}n - \frac{1}{k+1}$ for every pair of distinct vertices x, y with $xy \notin A(D)$ and $\delta^0(D) > \frac{k-1}{k+1}n + 2$. Then D contains the kth power of a Hamilton cycle.*

The sharpness of both conditions in Conjecture 5.2 follows from the constructions of Châu [2]. For the Ore-type condition, let $t \geq 2$, take the complete $(k+1)$-partite graph on $n = (k+1)(t+1) - 1$ vertices with one part of size t and the remaining k parts of size $t+1$. Its minimum degree sum over non-adjacent pairs is $2k(t+1) - 2$, which is one less than the integer bound required in Conjecture 5.2. Moreover, it contains no kth power of a Hamilton cycle, since $\alpha(C_n^k) = \lfloor n/(k+1) \rfloor = t$.

For the minimum semi-degree condition, for infinitely many admissible values of n, Châu constructs a graph G satisfying the Ore-type condition and containing a vertex v with $d_G(v) = \frac{k-1}{k+1}n + 2$. Here $G[N_G(v)]$ is obtained from a complete $(k-1)$-partite graph by adding edges within its unique larger part so that this part induces a disjoint union of triangles. In particular, $G[N_G(v)]$ contains no copy of P_{2k}^{k-1}. If G contained the kth power of a Hamilton cycle, then the k predecessors and the k successors of v on the underlying cycle would contain such a copy, a contradiction. Replacing every edge in these constructions by a pair of opposite arcs gives the corresponding symmetric digraphs. Thus the Ore-type bound cannot be lowered, and the strict minimum semi-degree condition in Conjecture 5.2 cannot be relaxed.

Higher powers of Hamilton cycles. Finally, Question 1.1 remains widely open for $k \geq 3$. While our almost covering and absorbing lemmas apply to all integers $k \geq 2$ under the condition $\delta^0(D) \geq \big(1 - 1/(k+2) + \gamma\big)n$, the main obstacle lies in the connecting step. In particular, one would need a suitable connecting lemma for k-paths when $k \geq 3$. More generally, determining the exact minimum semi-degree threshold for the kth power of a Hamilton cycle stands as a fascinating and challenging problem.

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Data availability

No data was used for the research described in the article.

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